

## 30.2 The elastic net

When several regressors are near copies of one another (repeated measurements, genes in one pathway), the lasso tends to keep one and drop the rest, and which one can change with a small perturbation of the data. With two identical columns its solution is not even unique (Exercise B1 of Section 27.4). Ridge spreads the coefficient evenly over the copies but never sets anything to zero. Zou and Hastie (2005) proposed using both penalties.

### 30.2.1 Definition and basic properties

#### Definition 30.2.1. Elastic net

For  $\lambda_1, \lambda_2 \geq 0$ , the **elastic net** criterion is

$$Q(\mathbf{b}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda_1 \|\mathbf{b}\|_1 + \frac{1}{2} \lambda_2 \|\mathbf{b}\|^2, \quad (30.2.1)$$

and its minimizer  $\hat{\boldsymbol{\beta}}^{\text{en}}$  is the (naive) **elastic net estimate**. An equivalent parameterization is  $\lambda_1 = \alpha\lambda$ ,  $\lambda_2 = (1 - \alpha)\lambda$ , with a mixing parameter  $\alpha \in [0, 1]$  and an overall level  $\lambda \geq 0$ .

With  $\lambda_2 = 0$  this is the lasso, and with  $\lambda_1 = 0$  ridge. Here the columns of  $\mathbf{X}$  are centred with unit length and  $\mathbf{y}$  is centred, so  $\rho_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$  is a sample correlation.

#### Proposition 30.2.1. Elastic net: uniqueness, augmented data, grouping

Let  $\lambda_2 > 0$ .

- (a) The criterion Eq. 30.2.1 has a unique minimizer, whatever the rank of  $\mathbf{X}$ . It is characterized by

$$\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) - \lambda_2 \hat{\beta}_j = \lambda_1 \text{sign}(\hat{\beta}_j) \quad (\hat{\beta}_j \neq 0), \quad |\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})| \leq \lambda_1 \quad (\hat{\beta}_j = 0). \quad (30.2.2)$$

- (b) (*Augmented data.*) Let  $\mathbf{X}^* = \begin{pmatrix} \mathbf{X} \\ \sqrt{\lambda_2} \mathbf{I}_p \end{pmatrix}$  and  $\mathbf{y}^* = \begin{pmatrix} \mathbf{y} \\ \mathbf{0} \end{pmatrix}$ . Then  $\hat{\boldsymbol{\beta}}^{\text{en}}$  is the lasso estimate, with parameter  $\lambda_1$ , for the data  $(\mathbf{X}^*, \mathbf{y}^*)$ . The matrix  $\mathbf{X}^*$  has full column rank  $p$ .
- (c) (*Orthonormal design.*) If  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ , then  $\hat{\beta}_j^{\text{en}} = \text{sign}(z_j)(|z_j| - \lambda_1)_+ / (1 + \lambda_2)$ , with  $\mathbf{z} = \mathbf{X}^\top \mathbf{y}$ .
- (d) (*Grouping effect.*) If  $\hat{\beta}_i \hat{\beta}_j > 0$ , then

$$|\hat{\beta}_i - \hat{\beta}_j| \leq \frac{\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\| \|\mathbf{x}_i - \mathbf{x}_j\|}{\lambda_2} \leq \frac{\|\mathbf{y}\| \sqrt{2(1 - \rho_{ij})}}{\lambda_2}. \quad (30.2.3)$$

- (e) (*Identical columns.*) If  $\mathbf{x}_i = \mathbf{x}_j$ , then  $\hat{\beta}_i = \hat{\beta}_j$ .

*Proof.*

- (a) The penalty  $\lambda_1 \|\mathbf{b}\|_1 + \frac{1}{2} \lambda_2 \|\mathbf{b}\|^2$  is strictly convex, because  $\|\mathbf{b}\|^2$  is strictly convex and  $\|\mathbf{b}\|_1$  is convex. It is at least  $\frac{1}{2} \lambda_2 \|\mathbf{b}\|^2$ , so  $Q \rightarrow \infty$ . Existence and uniqueness follow

as in Theorem 30.1.1(a) and (c). By Lemma 30.1.2 applied to the whole penalty, with the gradient  $\lambda_2 \hat{\boldsymbol{\beta}}$  of the smooth part moved to the left-hand side, and Lemma 30.1.3, the minimizer is characterized by Eq. 30.2.2.

- (b)  $\|\mathbf{y}^* - \mathbf{X}^* \mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2 + \lambda_2 \|\mathbf{b}\|^2$ . So  $\frac{1}{2} \|\mathbf{y}^* - \mathbf{X}^* \mathbf{b}\|^2 + \lambda_1 \|\mathbf{b}\|_1$  is Eq. 30.2.1. The last  $p$  rows of  $\mathbf{X}^*$  are  $\sqrt{\lambda_2} \mathbf{I}_p$ , which has rank  $p$ .
- (c) With  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ , Eq. 30.2.1 equals  $\sum_j \{\frac{1}{2}(1 + \lambda_2)b_j^2 - z_j b_j + \lambda_1 |b_j|\}$  plus a constant. Dividing the  $j$ th term by  $1 + \lambda_2$  gives  $\frac{1}{2}\{b_j - z_j/(1 + \lambda_2)\}^2 + \{\lambda_1/(1 + \lambda_2)\}|b_j|$  plus a constant. By the univariate lasso solution (Theorem 27.4.2), this is minimized by soft thresholding  $z_j/(1 + \lambda_2)$  at  $\lambda_1/(1 + \lambda_2)$ , which is the stated formula.
- (d) Both coefficients are nonzero with a common sign  $s$ , so Eq. 30.2.2 gives  $\mathbf{x}_i^\top \mathbf{r} - \lambda_2 \hat{\beta}_i = \lambda_1 s = \mathbf{x}_j^\top \mathbf{r} - \lambda_2 \hat{\beta}_j$ , where  $\mathbf{r} = \mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}$ . Subtracting,  $\lambda_2(\hat{\beta}_i - \hat{\beta}_j) = (\mathbf{x}_i - \mathbf{x}_j)^\top \mathbf{r}$ , and Cauchy–Schwarz gives the first inequality. For the second,  $\|\mathbf{x}_i - \mathbf{x}_j\|^2 = 2 - 2\rho_{ij}$  for unit-length columns, and  $\frac{1}{2}\|\mathbf{r}\|^2 \leq Q(\hat{\boldsymbol{\beta}}) \leq Q(\mathbf{0}) = \frac{1}{2}\|\mathbf{y}\|^2$ .
- (e) Swapping coordinates  $i$  and  $j$  of  $\hat{\boldsymbol{\beta}}$  changes neither  $\mathbf{X} \hat{\boldsymbol{\beta}}$  nor the penalty. So the swapped vector is also a minimizer, and by uniqueness it equals  $\hat{\boldsymbol{\beta}}$ .

■

By (b), the elastic net is a lasso on  $n + p$  observations with a full-rank design, so it can select all  $p$  regressors even when  $p > n$ ; a unique lasso solution has at most  $\min(n, p)$  nonzero coefficients (Exercise C1). Part (d) is the grouping effect of Zou and Hastie (2005): as  $\rho_{ij} \rightarrow 1$ , coefficients of the same sign are forced together at a rate set by  $\lambda_2$ . No such bound holds for the lasso.

### Example. Three near-copies of one signal

The script simulates  $n = 50$  observations of  $p = 10$  regressors. The first three are noisy copies of a common variable, with pairwise correlations between 0.93 and 0.95, and each has coefficient 2 (on the unit-length scale). The other seven are independent noise. With  $\lambda_1 = 0.8$ :

|                                    | $\hat{\beta}_1$ | $\hat{\beta}_2$ | $\hat{\beta}_3$ | nonzero |
|------------------------------------|-----------------|-----------------|-----------------|---------|
| lasso ( $\lambda_2 = 0$ )          | 2.335           | 0.000           | 2.522           | 2       |
| elastic net<br>( $\lambda_2 = 1$ ) | 1.248           | 1.103           | 1.236           | 4       |
| rescaled elastic<br>net            | 2.496           | 2.207           | 2.471           | 4       |

The lasso drops the second copy. The elastic net keeps all three with similar coefficients (and one noise regressor, at 0.081). For the pair (1, 2), with correlation 0.927, the spread 0.144 is well inside Eq. 30.2.3, whose middle term is 1.577 (residual norm 4.129) and right-hand side 2.532 ( $\|\mathbf{y}\| = 6.630$ ). Along the lasso path (Figure 30.2.1) the second copy enters only after  $\lambda$  has fallen by a factor of about 26; along the elastic net path all three enter at the same grid value.

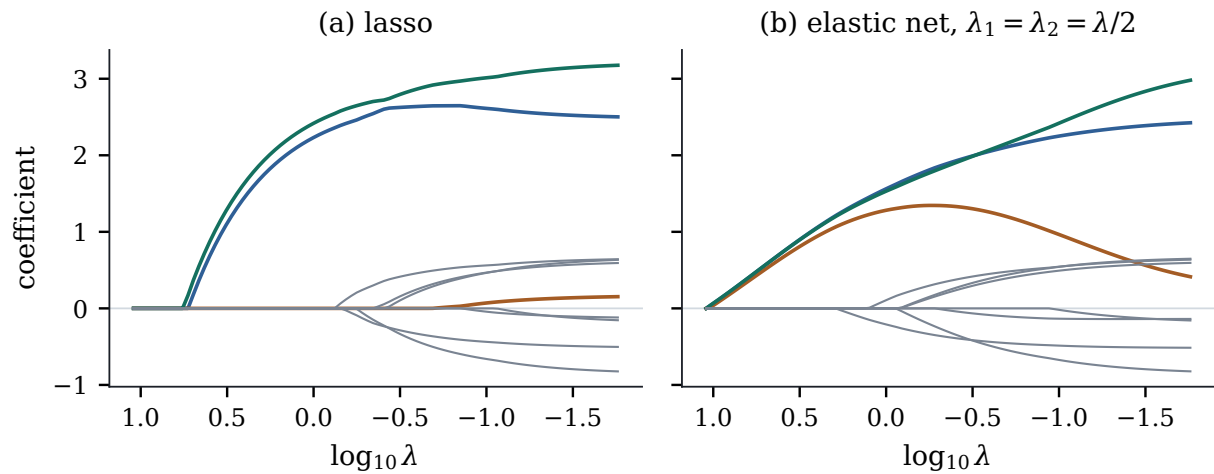


Figure 30.2.1: Coefficient paths for the data of [Example \(Three near-copies of one signal\)](#), plotted against  $\log_{10} \lambda$ , with  $\lambda$  decreasing to the right. The three correlated regressors are coloured and the seven noise regressors are grey. (a) The lasso. (b) The elastic net with  $\alpha = \frac{1}{2}$ , that is,  $\lambda_1 = \lambda_2 = \lambda/2$ .

## 30.2.2 Computing it: coordinate descent

**Cyclic coordinate descent**, used for the lasso in Lemma 27.4.4, minimizes over one coordinate at a time, the others fixed, and sweeps until nothing changes.

### Proposition 30.2.2. Coordinate update

Fix all coordinates of  $\mathbf{b}$  except  $b_j$ , and let  $\mathbf{r}_{(j)} = \mathbf{y} - \sum_{k \neq j} \mathbf{x}_k b_k$  be the partial residual. The value of  $b_j$  that minimizes [Eq. 30.2.1](#) is

$$b_j = \frac{\text{sign}(\mathbf{x}_j^\top \mathbf{r}_{(j)}) (|\mathbf{x}_j^\top \mathbf{r}_{(j)}| - \lambda_1)_+}{\|\mathbf{x}_j\|^2 + \lambda_2}. \quad (30.2.4)$$

*Proof.* As a function of  $b_j$  alone, [Eq. 30.2.1](#) equals  $\frac{1}{2}(\|\mathbf{x}_j\|^2 + \lambda_2)b_j^2 - b_j \mathbf{x}_j^\top \mathbf{r}_{(j)} + \lambda_1 |b_j|$  plus a constant. Divided by  $\|\mathbf{x}_j\|^2 + \lambda_2$ , this is a univariate lasso problem of the form in the proof of [Proposition 30.2.1\(c\)](#). Its solution is [Eq. 30.2.4](#). ■

Because the nonsmooth part is separable, every limit point of the iterates is a minimizer (Tseng, 2001; cited, not proved), so the iterates converge when the minimizer is unique, as for the elastic net with  $\lambda_2 > 0$ . For a coupling penalty such as  $\sum_j |b_j - b_{j-1}|$ , coordinate descent can stall. Whole paths are computed on a decreasing grid of  $\lambda$ , each fit starting from the last (Friedman, Hastie, Höfling and Tibshirani, 2007; Friedman, Hastie and Tibshirani, 2010). The listing works through the Gram matrix, as the next proposition requires.

```
import numpy as np

def soft(z, lam):
    return np.sign(z) * np.maximum(np.abs(z) - lam, 0.0)
```

```

def cd_quadratic(G, c, lam1, b=None, tol=1e-12, max_sweeps=100000):
    """Minimize (1/2) b'Gb - c'b + lam1 ||b||_1 by cyclic coordinate descent."""
    p = len(c)
    b = np.zeros(p) if b is None else b.copy()
    for _ in range(max_sweeps):
        largest = 0.0
        for j in range(p):
            old = b[j]
            partial = c[j] - G[j] @ b + G[j, j] * old      # c_j - sum_{k != j} G_jk
            b_k
            b[j] = soft(partial, lam1) / G[j, j]
            largest = max(largest, abs(b[j] - old))
        if largest < tol:
            return b
    raise RuntimeError("coordinate descent did not converge")

def elastic_net(X, y, lam1, lam2, b=None):
    """The (naive) elastic net: (1/2)||y - Xb||^2 + lam1 ||b||_1 + (lam2/2)||b||^2.
    """
    G = X.T @ X + lam2 * np.eye(X.shape[1])
    return cd_quadratic(G, X.T @ y, lam1, b)

rng = np.random.default_rng(3002)
n, p = 50, 10
z = rng.normal(size=n)
X = rng.normal(size=(n, p))
X[:, :3] = z[:, None] + 0.3 * rng.normal(size=(n, 3))    # three near-copies of one
                  signal
X -= X.mean(axis=0)
X /= np.linalg.norm(X, axis=0)                          # centred, unit-length
                  columns
beta = np.r_[2.0, 2.0, 2.0, np.zeros(p - 3)]
y = X @ beta + 0.5 * rng.normal(size=n)
y -= y.mean()

lam1 = 0.8
b_lasso = elastic_net(X, y, lam1, 0.0)
b_naive = elastic_net(X, y, lam1, 1.0)
print("lasso", np.round(b_lasso[:5], 3))
print("naive EN", np.round(b_naive[:5], 3))
print("rescaled EN", np.round(2.0 * b_naive[:5], 3))    # (1 + lam2) * naive

```

[Run this code online](#)

### 30.2.3 Double shrinkage and the rescaled elastic net

By [Proposition 30.2.1\(c\)](#), under an orthonormal design the elastic net soft-thresholds and then divides by  $1 + \lambda_2$ : surviving coefficients are shrunk twice. Zou and Hastie (2005) proposed reporting  $(1 + \lambda_2)\hat{\beta}^{\text{en}}$  instead, which in general is a lasso with a modified Gram matrix.

#### Proposition 30.2.3. The rescaled elastic net

Let the columns of  $\mathbf{X}$  have unit length and let  $\lambda_2 > 0$ . Then  $\tilde{\beta} = (1 + \lambda_2)\hat{\beta}^{\text{en}}$  is the unique

minimizer of

$$\frac{1}{2} \mathbf{c}^\top \boldsymbol{\Sigma}_{\lambda_2} \mathbf{c} - \mathbf{y}^\top \mathbf{X} \mathbf{c} + \lambda_1 \|\mathbf{c}\|_1, \quad \boldsymbol{\Sigma}_{\lambda_2} = \frac{\mathbf{X}^\top \mathbf{X} + \lambda_2 \mathbf{I}}{1 + \lambda_2}. \quad (30.2.5)$$

*Proof.* Up to the constant  $\frac{1}{2} \mathbf{y}^\top \mathbf{y}$ , Eq. 30.2.1 is  $\frac{1}{2} \mathbf{b}^\top (\mathbf{X}^\top \mathbf{X} + \lambda_2 \mathbf{I}) \mathbf{b} - \mathbf{y}^\top \mathbf{X} \mathbf{b} + \lambda_1 \|\mathbf{b}\|_1$ . Substitute  $\mathbf{b} = \mathbf{c}/(1 + \lambda_2)$  and multiply by  $1 + \lambda_2 > 0$ . The result is Eq. 30.2.5. The substitution is a bijection, so minimizers correspond, and the minimizer of Eq. 30.2.1 is unique by Proposition 30.2.1(a). ■

The lasso has  $\mathbf{X}^\top \mathbf{X}$  in place of  $\boldsymbol{\Sigma}_{\lambda_2}$ , whose correlations are divided by  $1 + \lambda_2$ : the rescaled elastic net is a lasso computed as if the regressors were less correlated. The rescaling is a heuristic; in Example (Three near-copies of one signal), where the true coefficients are 2, the naive estimates are about half that and the rescaled ones somewhat too large. The glmnet software (Friedman, Hastie and Tibshirani, 2010) fits the naive criterion.

## 30.2.4 Exercises

### 30.2.4.1 A. Check your understanding

#### Exercise. A1

Under an orthonormal design with  $\mathbf{z} = \mathbf{X}^\top \mathbf{y} = (3, 0.5, -1.2)^\top$ ,  $\lambda_1 = 1$  and  $\lambda_2 = 1$ , compute the lasso, ridge, naive elastic net and rescaled elastic net estimates.

*Solution.* Lasso ( $\lambda = 1$ ):  $(2, 0, -0.2)$ . Ridge ( $\lambda = 1$ ):  $\mathbf{z}/2 = (1.5, 0.25, -0.6)$ . Naive elastic net: the lasso divided by 2,  $(1, 0, -0.1)$ . Rescaled: multiply by  $1 + \lambda_2 = 2$ , which gives the lasso,  $(2, 0, -0.2)$ .

### 30.2.4.2 B. Practice

#### Exercise. B1

Show that if  $\hat{\beta}_i \hat{\beta}_j < 0$ , then  $|\hat{\beta}_i + \hat{\beta}_j| \leq \|\mathbf{y}\| \sqrt{2(1 + \rho_{ij})}/\lambda_2$ . Interpret this for two regressors with correlation close to  $-1$ .

*Solution.* With signs  $s$  and  $-s$ , Eq. 30.2.2 gives  $\mathbf{x}_i^\top \mathbf{r} - \lambda_2 \hat{\beta}_i = \lambda_1 s$  and  $\mathbf{x}_j^\top \mathbf{r} - \lambda_2 \hat{\beta}_j = -\lambda_1 s$ . Adding,  $\lambda_2(\hat{\beta}_i + \hat{\beta}_j) = (\mathbf{x}_i + \mathbf{x}_j)^\top \mathbf{r}$ , and  $\|\mathbf{x}_i + \mathbf{x}_j\|^2 = 2 + 2\rho_{ij}$ . Bound  $\|\mathbf{r}\| \leq \|\mathbf{y}\|$  as before. If  $\rho_{ij} \approx -1$ , then  $\mathbf{x}_j \approx -\mathbf{x}_i$ , and the bound forces  $\hat{\beta}_j \approx -\hat{\beta}_i$ . The two regressors are treated as one regressor with a sign flip.

**Exercise. B2**

Let  $\mathbf{X} = [\mathbf{x}, \mathbf{x}]$  with  $\|\mathbf{x}\| = 1$ . Find the elastic net estimate in closed form and compare it with the ridge and lasso estimates.

**30.2.4.3 C. Going deeper****Exercise. C1**

Suppose  $p > n$  and the lasso estimate  $\hat{\boldsymbol{\beta}}$  is unique with active set  $A$ . Use the KKT conditions to show that the columns  $\mathbf{X}_A$  are linearly independent, and deduce that  $|A| \leq n$ . (Hint: if  $\mathbf{X}_A \mathbf{d} = \mathbf{0}$  with  $\mathbf{d} \neq \mathbf{0}$ , move along  $\hat{\boldsymbol{\beta}} + \tau(\mathbf{d}, \mathbf{0})$  and use that the fit and the  $\ell_1$  norm are constant among minimizers.)